**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 046 · Lesson 046 · Objects in Arrays and Lists

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What is a constructor used for?

Q2. What does private mean for an instance field?

Q3. What index identifies the first C# array item?

Q4. 031

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

### Key vocabulary

object — Instance of a class.

reference — Value identifying an object instance.

aliasing — Several references to the same object.

index — Position in a sequence.

### New learning

Observe — A class describes an object; the array stores references to two Book instances.

Explain — Indexes start at 0. shelf[1] identifies the second reference, then .Title reads its property.

Apply — An array has a fixed length after creation; individual array elements can refer to objects whose properties change.

### Worked example

Trace the title stored at books[0] in a two-object Book array.

1. Trace and justify — Index 0 refers to the first object; its Title is Ada.

# We Do · Practise with guidance

In a different array, which title prints after updating the object at index 1?

```
Book[] b={new Book{Title="One"},new Book{Title="Two"}};
b[1].Title="Updated";
Console.WriteLine(b[1].Title);
```

Trace and justify

Index 1 refers to the second object. The property update changes its Title; output is Updated.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Given shelf titles Dune,Holes, give output of shelf[1].Title and reason. [4]

### Question 2

Write C# to create an array of two Book objects with titles A and B, then print first title. [6]

### Question 3

Two array elements reference the SAME Book object. Changing its title via first reference is then printed via second. Explain. [6]

## Exit Ticket · Answer before checking

Updating a property through one reference can be visible through another alias.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Initialising an object's state.

Q2. It cannot be accessed directly from outside the class.

Q3. Zero.

Q4. A rising-edge D flip-flop has Q=0 and D=1 just before the next active edge. Give Q after it and why.

### You Do pages · Independent questions

Q1. Holes [2]; 0-based index 1 selects second element and its Title [2].

Q2. Book[] b={new Book{Title="A"},new Book{Title="B"}}; [4] Console.WriteLine(b[0].Title); [2], assuming Book declares Title property.

Q3. Both elements hold references [2] to same instance [2]; mutation visible through either reference [2]; this is aliasing, not array-index copying.

### Exit Ticket · Model answer

True. Both can point to the same instance.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**